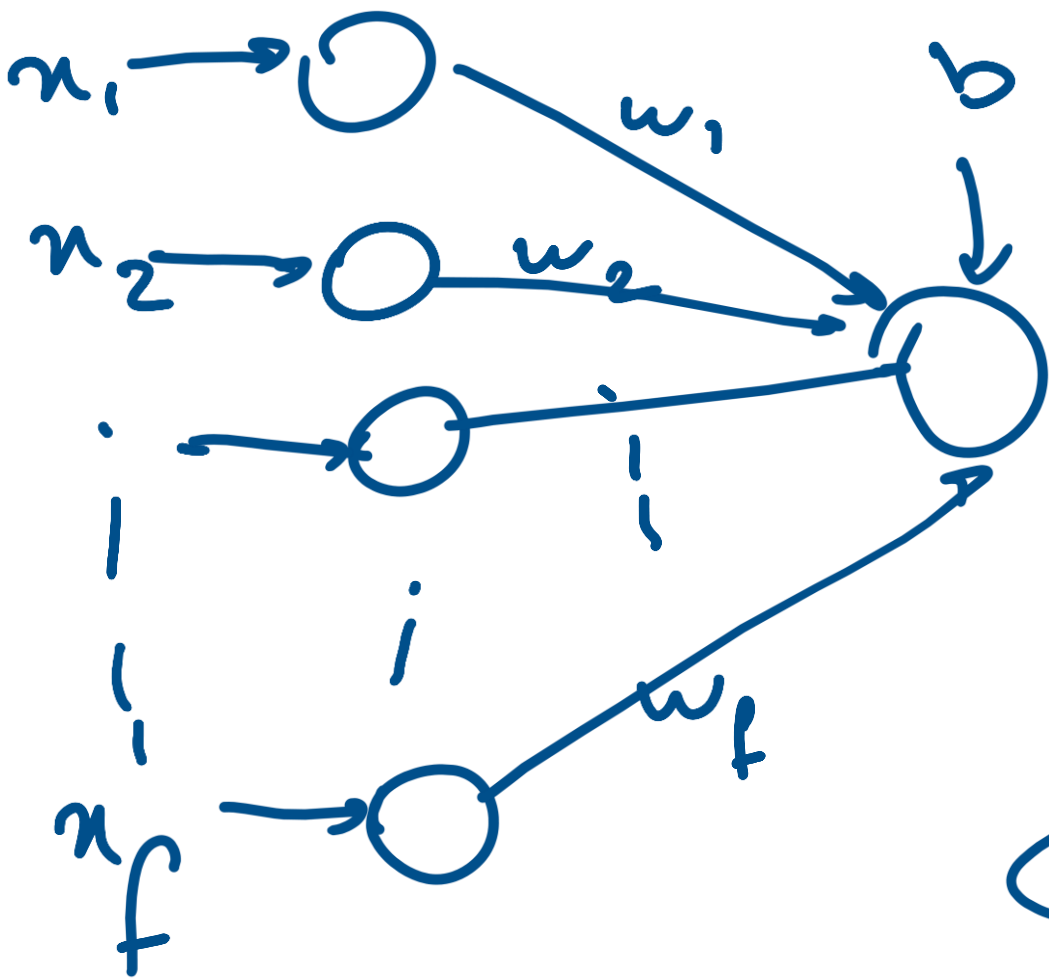
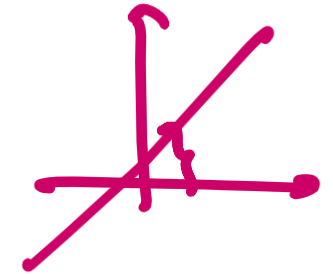


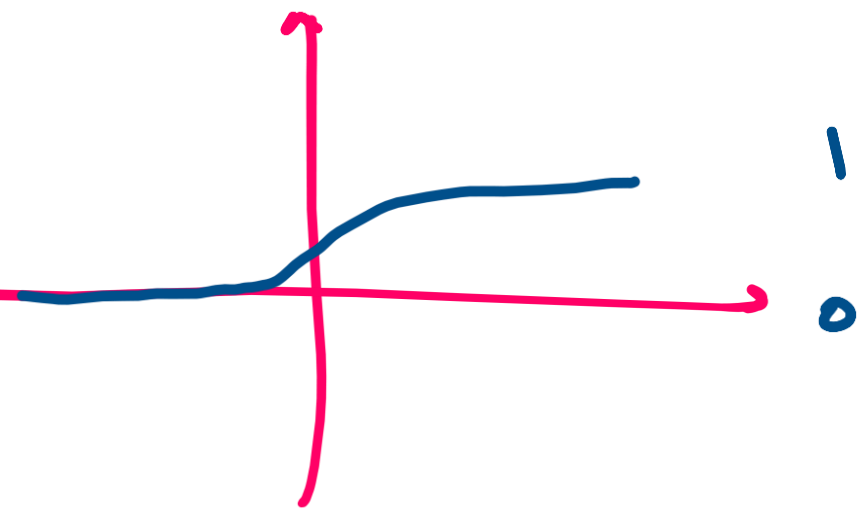


$$y = \sum_{i=1}^f w_i x_i + b$$



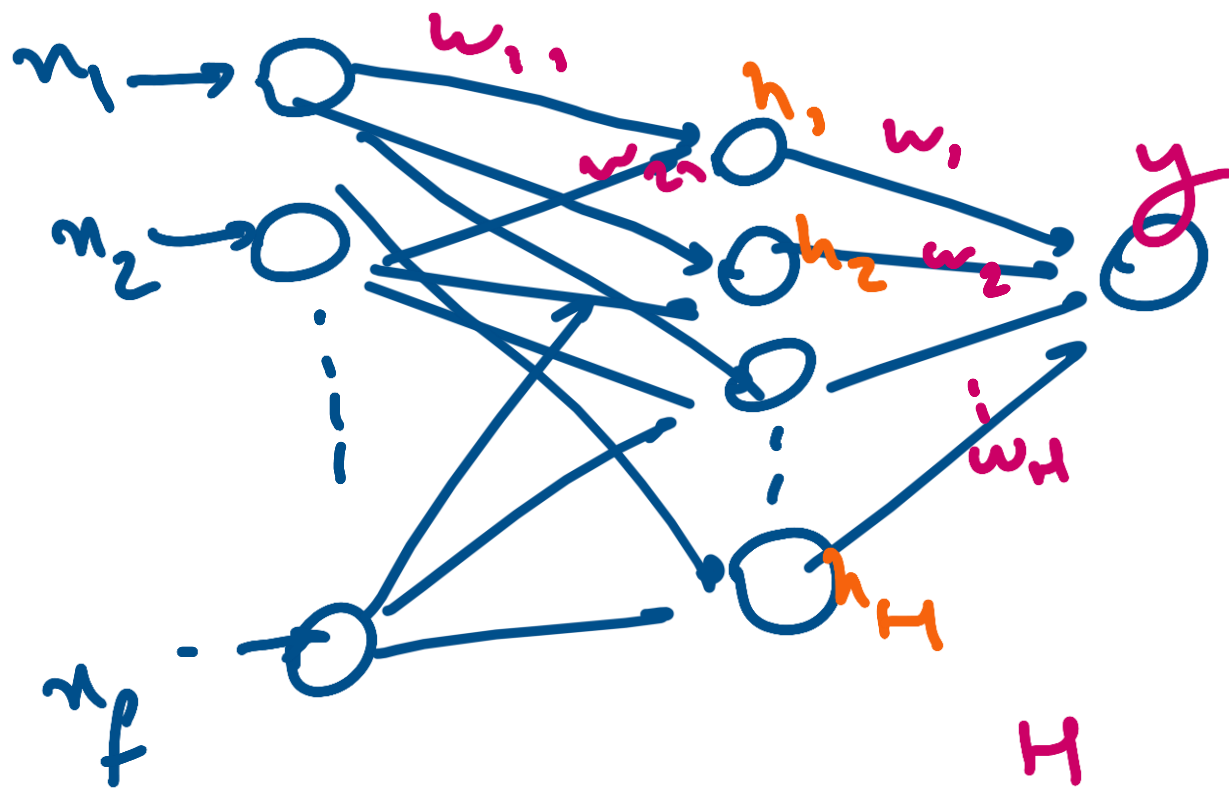
$$y = \text{logistic} \left(\sum_{i=0}^f w_i x_i \right)$$

$w_0 \rightarrow b$
 $x_0 \rightarrow 1$



$$\text{Logistic Sigmoid}(x) = \frac{1}{1 + e^{-x}}$$

MLP



$$h_i = w_{i1}x_1 + w_{i2}x_2 + \dots + w_{ip}x_p + w_{i0}$$

$$h_i = \text{Af}_i \left(\sum_{i=0}^p w_{i1}x_i \right)$$

$$h_j = \text{Af}_j \left(\sum_{i=0}^p w_{ij}x_i \right)$$

$$y = \text{Af}_y \left(\sum_{j=0}^H w_{jy}h_j \right)$$

4 class

y_1	y_2	y_3	y_4	
0	0	0	1	$\rightarrow C_1$
0	0	1	0	C_2
0	1	0	0	C_3
1	0	0	0	C_4

$A_{f_2} \rightarrow \text{Softmax}$

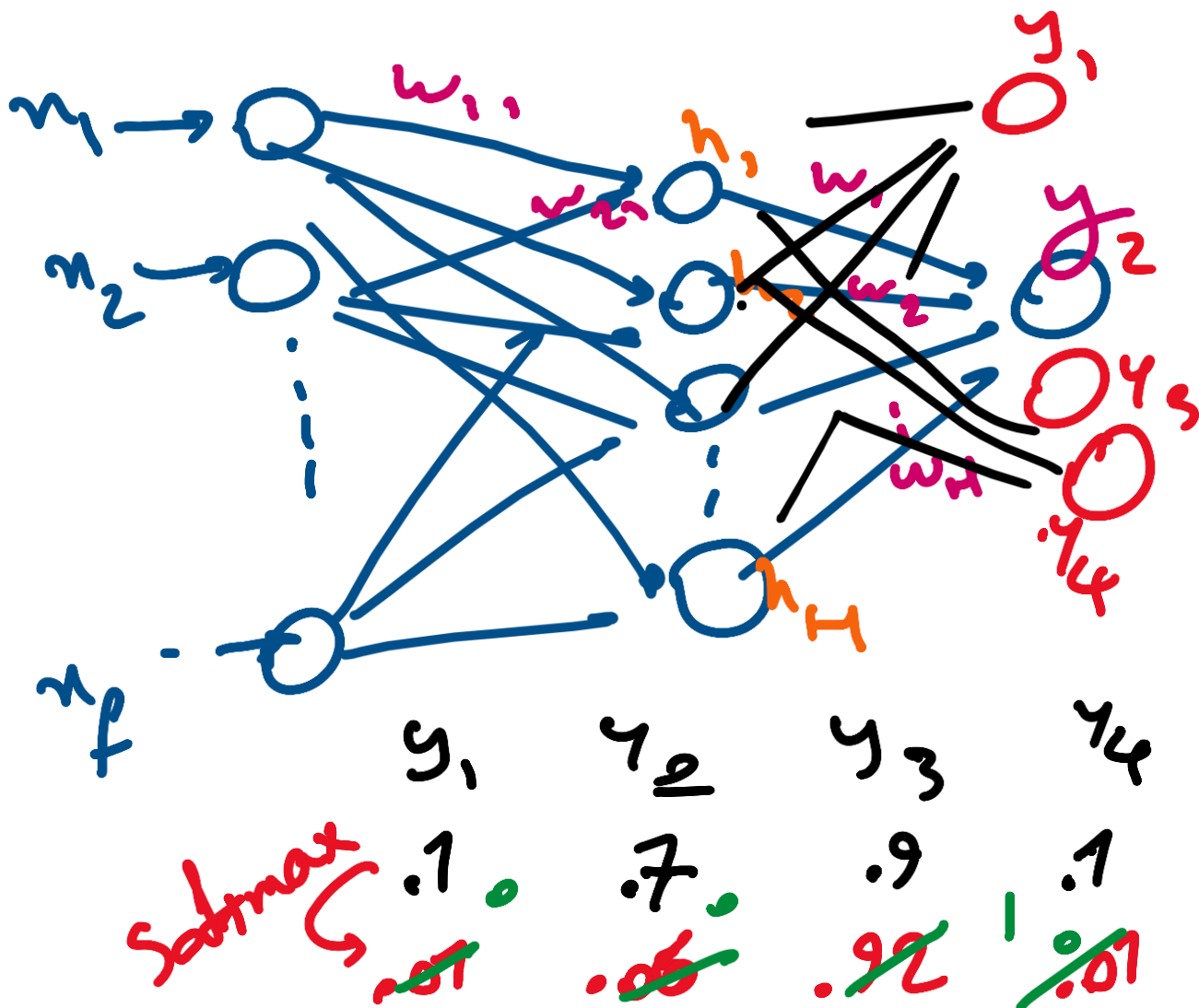
$$\text{Softmax}(y_i) = \frac{e^{y_i}}{\sum_{j=1}^C e^{y_j}}$$

Handwritten formula for Softmax. The denominator is annotated with a 'C' and an arrow pointing to the summation symbol, indicating the number of classes.

2 2 2 3

MLP

$$h_i = w_{i1}x_1 + w_{i2}x_2 + \dots + w_{if}x_f + w_{i0}$$



y_1	y_2	y_3	y_4	
1	0	0	0	$\rightarrow c_1$
0	0	0	1	$\rightarrow c_4$
1	1	0	1	$\rightarrow c_3$
0	0	1	0	$\rightarrow c_3$
1	0	0	1	$\rightarrow c_4$
0	0	0	1	$\rightarrow c_4$

$$\text{GD: } w_N = w_0 - \eta \frac{\partial L}{\partial w}$$

SGD

Adam
Loss → Classification

BCE
CCE + $\alpha ||w||$

→ Regression

MSE + $\alpha ||w||$

Regression	y_{GT}	2	3	2.7		1	2	2
		1	1	1		1	5	3
	y_0	2.1	3.5	?		1	5	3

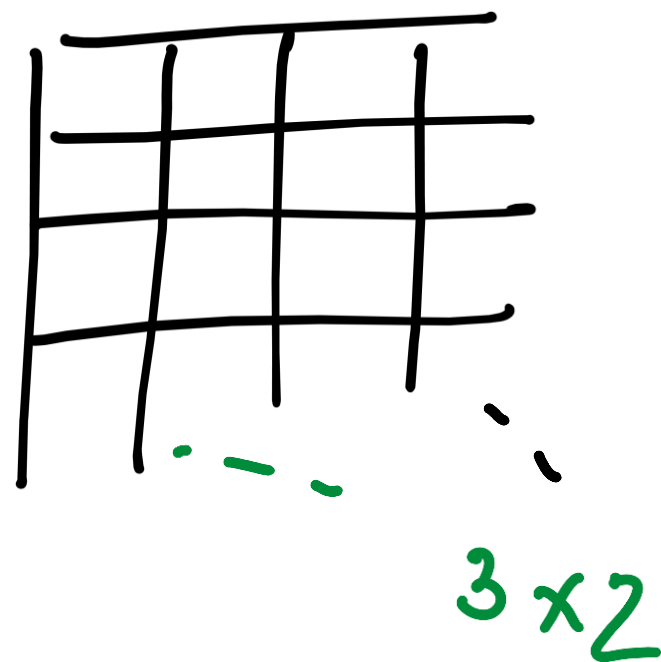
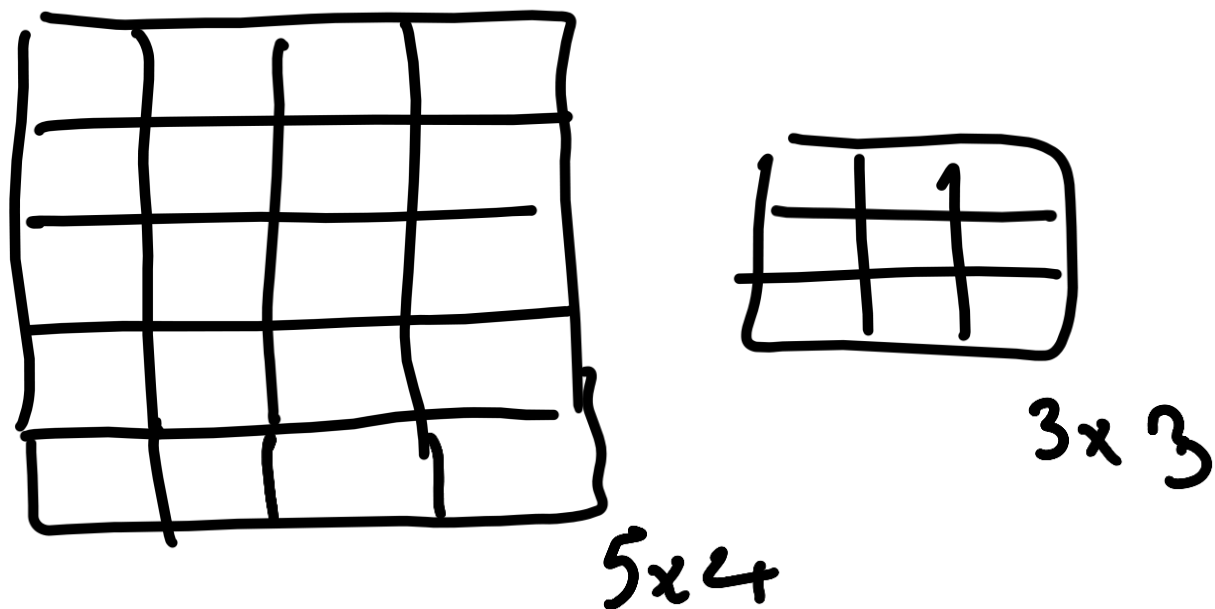
Conv.

$$f \star g(t) = \int_{-\infty}^{\infty} f(\tau) g(t-\tau) d\tau$$

$$f = [3, 4, 7, 6, 9, 2, 5, 1]$$

$$g = [1, 0, 1]$$

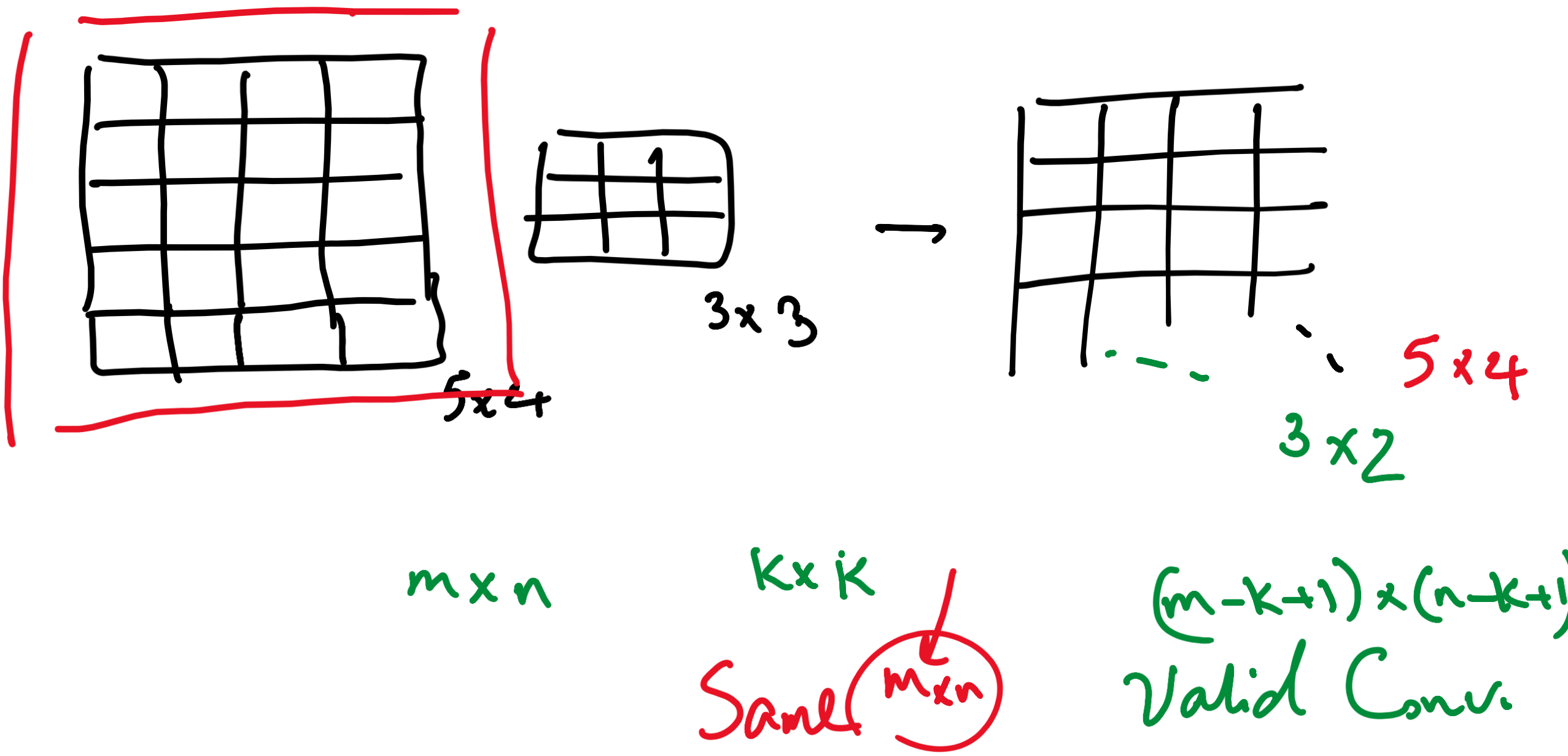
$$f * g = [10, 10, 16, 8, 14, 3]$$



$m \times n$

$K \times K$

$(m - K + 1) \times (n - K + 1)$
Valid Conv.



1. Bernoulli Distribution

The **Bernoulli distribution** models a random variable that takes values in $\{0, 1\}$, with probability p of being 1:

$$P(y \mid p) = p^y(1 - p)^{1-y}, \quad y \in \{0, 1\}, \quad 0 \leq p \leq 1$$

- If $y = 1$: $P(y = 1) = p$
- If $y = 0$: $P(y = 0) = 1 - p$

This is the foundation for binary classification.

$$L = \prod_{i=1}^N \hat{p}_i^{y_i} (1 - \hat{p}_i)^{1-y_i}$$

negative log-likelihood (NLL):

$$\ell = - \sum_{i=1}^N \left[y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i) \right]$$

For K -class classification:

- Label y is a **one-hot vector** $y \in \{0, 1\}^K$, where only one element is 1.
- Model outputs a probability distribution $\hat{p} = \text{softmax}(z)$:

$$\hat{p}_k = \frac{e^{z_k}}{\sum_{j=1}^K e^{z_j}}$$

The categorical distribution:

$$P(y \mid \hat{p}) = \prod_{k=1}^K \hat{p}_k^{y_k}$$

Taking NLL:

$$\ell = - \sum_{k=1}^K y_k \log \hat{p}_k$$

1. Entropy in Information Theory

For a true distribution $P(y)$, the **entropy** is:

$$H(P) = - \sum_y P(y) \log P(y)$$

This measures the **uncertainty** (average information content) of the distribution.

- If outcomes are very certain, entropy is low.
- If outcomes are very unpredictable, entropy is high.

2. Cross-Entropy

If we approximate $P(y)$ (the true distribution) with a model distribution $Q(y)$, the **cross-entropy** is:

$$H(P, Q) = - \sum_y P(y) \log Q(y)$$

This measures how many bits, on average, are needed to encode samples from P if we use the wrong distribution Q .

3. KL Divergence (the connection)

Cross-entropy can be written as:

$$H(P, Q) = H(P) + D_{KL}(P \| Q)$$

where $D_{KL}(P \| Q)$ is the **Kullback–Leibler divergence** (extra cost of using Q instead of the true P).

- Since $H(P)$ is fixed (depends only on the data, not the model), minimizing cross-entropy is **equivalent to minimizing KL divergence** between the true distribution and our model's distribution.

